

PrediCare

AI-Powered District Health Operating System

Predict operational failures at PHCs and CHCs before patients are affected — close the intervention loop in under 5 minutes.

Purpose of this document: source material for building a 10–12 slide pitch deck. Each section below maps to one or more slides — organized by problem, solution, AI, users, deployability, and evidence of a working product. Pull quotes and numbers directly; trim prose for slide bullets.

1. The Problem

India's Primary Health Centres (PHCs) and Community Health Centres (CHCs) — the frontline of rural and semi-urban healthcare — still track medicine stock, patient footfall, bed availability, doctor attendance, and diagnostic supplies largely on paper or in disconnected local registers.

The result: district health administrators discover operational failures **2–5 days after they occur** — a medicine stockout, an absent duty doctor, a diagnostic kit shortage — by which point patients have already been turned away or under-treated.

- No early-warning signal for stockouts, staffing gaps, or capacity overruns
- No visibility across facilities — a district officer cannot see which of 50+ PHCs need help right now
- No mechanism to redistribute surplus stock from a well-stocked facility to a failing one before it's too late
- Language and connectivity barriers exclude frontline workers from any existing digital tooling

Slide suggestion: 1 slide, visual-first. A single stat or before/after (e.g. "2–5 days to discover a stockout" vs "under 5 minutes with PrediCare") lands harder than a bullet list.

2. The Solution

PrediCare is a predictive control tower that sits above the daily operations of a health district. It ingests routine facility data (stock levels, footfall, bed occupancy, staff attendance, diagnostic availability) and turns it into forward-looking, actionable intelligence — then closes the loop by letting officers act in one tap, and field workers get instructions over WhatsApp in their own language.

What it does

- **Forecasts stockouts 48–72 hours ahead**, with a confidence score and a plain-language reason ("Insulin out in 2 days — monsoon demand +20%, dengue trend +15%")
- **Predicts patient demand** (footfall, staffing needs) using weather, disease-season, and calendar signals
- **Recommends inter-facility redistribution** before a shortage hits — "Transfer 60 units from PHC-8, 4km away, saves ₹18,000" — computed via an optimization solver, not a rule of thumb
- **Scores every facility 0–100** across five dimensions (medicine, doctor attendance, beds, wait time, diagnostics) and continuously surfaces the facilities most in need of intervention
- **Closes the loop over WhatsApp** — a district officer approves or defers a redistribution plan with one tap; the receiving/sending facility's field worker gets a WhatsApp instruction in their own language. No

dashboard login required for the field side.

- **Works offline-first** for field workers (up to 72 hours of local data entry with automatic sync once connectivity returns) and in multiple regional languages with voice input, so low-connectivity, low-literacy contexts aren't excluded

The core insight: the data to predict these failures already exists inside every PHC/CHC's daily routine — stock counts, patient logs, attendance registers. Nobody needs to collect anything new. PrediCare's job is to read the signal that's already there, days before a human would notice it manually.

3. Who It Serves — Three Roles, One System

Role	What they see / do	Interface
District Officer	District-wide live dashboard: map of every facility color-coded by health, active alert feed, bottom-5 at-risk facilities, one-tap approve/defer on redistribution plans, AI assistant for natural-language queries over live data	Web dashboard (desktop-first, 10 languages)
PHC/CHC Admin (facility in-charge)	Single-facility view: their own health score, stock, bed matrix, test availability, active alerts, and any redistribution transfers touching their facility — read-only, scoped strictly to their own facility	Web dashboard, single-facility view
Field Worker	Daily data entry: patient count, footfall tally by type, geofenced check-in, doctor attendance, bed matrix, diagnostic test availability, medicine stock — voice input supported, works fully offline	Installable mobile web app (PWA), 7 languages

All three interact with the same underlying live data — a field worker's stock entry can trigger a stockout prediction within the hour, which can trigger a redistribution suggestion to a district officer, who approves it, which sends a WhatsApp instruction back to a field worker at another facility. One loop, three roles, minutes not days.

4. How the AI Works

Module	Model / Technique	Output
Stockout Prediction	Prophet + XGBoost time-series forecasting	Days until stockout, confidence score, plain-language reasoning
Diagnostics Shortage	Prophet + XGBoost	Reagent/test-kit runout forecast
Footfall Forecast	LightGBM	Expected patient volume + staffing suggestion
Redistribution Planning	OR-Tools linear optimization	Optimal transfer plan (source, destination, quantity) with cost/distance tradeoff
Facility Health Score	Weighted composite (medicine 25%, doctor 20%, beds 20%, wait-time 20%, diagnostics 15%)	0–100 score + Green/Yellow/Red status, updated continuously

Anomaly Detection	Isolation Forest	Outbreak signals / data-entry anomalies flagged for review
Conversational Assistant	Gemini (Google AI, multilingual)	Natural-language Q&A over live district data — voice-to-voice supported

This is not a single chatbot wrapper — each prediction is produced by a purpose-built model trained for its specific task, then surfaced through one unified alerting and scoring layer so a district officer sees one coherent picture, not seven dashboards.

Slide suggestion: this table works almost as-is on one slide. Consider leading with the Health Score and Redistribution rows since those are the most demo-able / visually intuitive; the LLM assistant is the "wow" close.

5. Inclusivity & Accessibility

- **10 languages on the district dashboard:** English, Hindi, Marathi, Gujarati, Punjabi, Tamil, Malayalam, Telugu, Kannada, Bengali
- **7 languages on the field-worker app:** English, Hindi, Marathi, Gujarati, Kannada, Tamil, Malayalam
- **Voice input** for field data entry — a worker can speak a patient count instead of typing, critical for low-literacy or time-pressured contexts
- **Guided in-app onboarding** — a "How to Use" walkthrough auto-shows on first login for both the field app and the district dashboard (including a voice-narrated navigation tour), so no separate training session is required to adopt the tool
- **WhatsApp-first field communication** — redistribution instructions and stockout alerts reach field workers over WhatsApp, a channel already in universal use, rather than requiring a new app login
- **Offline-first field app** — built as an installable Progressive Web App with local storage; a field worker can keep entering data for up to 72 hours without connectivity, with automatic sync once back online

6. Why It's Deployable

Built and running on Google Cloud today

This is not a local prototype — PrediCare is deployed and live on Google Cloud Platform + Firebase Hosting, using a fully managed, horizontally scalable cloud architecture:

Layer	Google Cloud Service
API / backend compute	Cloud Run (auto-scales to zero when idle, scales out under load)
Background job processing	Cloud Run (Celery worker + scheduler, always-on)
Primary database	Cloud SQL for PostgreSQL + PostGIS (geospatial facility queries)
Cache / task queue broker	Redis on Compute Engine, reachable only over a private VPC connector — never exposed to the public internet
Container images	Artifact Registry

Secrets	Secret Manager (DB credentials, JWT signing key, etc. — never hardcoded)
Frontend hosting	Firebase Hosting (dashboard + field app, both served over HTTPS/CDN)
Conversational AI	Gemini (Google AI)

This means PrediCare can go from a 10-facility pilot district to a state-wide rollout without a re-architecture — Cloud Run scales horizontally per request load, and Cloud SQL/Redis capacity can be increased independently of application code.

Designed for real-world PHC/CHC constraints

- **Low-bandwidth tolerant** — the field app is offline-first by design, not as an afterthought
- **No new hardware required** — runs in any modern mobile browser; the field app installs like a native app (PWA) without an app-store dependency
- **Containerized** — the entire backend ships as Docker images, portable to any Kubernetes-compatible infrastructure, including government-approved cloud (e.g. NIC/MeitY-empanelled clouds), not locked to a single vendor's proprietary services beyond standard Postgres/Redis/containers
- **Data-privacy conscious** — aggregated operational counts only; no individual patient PII is stored, which keeps the system aligned with DPDP Act 2023 principles

Deployed, working links (for the deck's "deployable" slide — screenshot or QR-code these):

District Dashboard: <https://predicare-dashboard.web.app>

Field Worker App: <https://predicare-field-app.web.app>

Backend API: <https://predicare-api-2vlcd6olsq-el.a.run.app>

7. Live Demo Script (3 Minutes)

1. Open the district dashboard → one facility shows RED on the map
2. Click into it → the AI explains: *"Insulin out in 2 days. Monsoon demand +20%, dengue trend +15%."*
3. The system proposes: *"Transfer 60 units from a facility 4km away. Saves ₹18,000."*
4. District officer taps **Approve** → the receiving facility's field worker gets a WhatsApp message in their own language
5. The facility's health score updates live: RED → YELLOW
6. Voice demo: a field worker says a patient count aloud in Hindi → the dashboard updates in real time

This loop — detection, explanation, recommendation, one-tap action, resolution — is the entire pitch in miniature. It's worth building the whole deck's narrative arc around this sequence.

8. Impact Potential

2–5 days → <5 min

time to detect an operational failure

48–72 hrs

advance stockout warning window

10 → thousands

facilities the same architecture scales to without re-engineering

- Every facility scored continuously — district officers stop reacting to complaints and start intervening before failure
- Redistribution turns siloed facility inventories into a shared district-wide buffer, cutting emergency procurement and stockout-driven patient turn-away
- WhatsApp-based closure means adoption doesn't require field workers to learn a new app or the officer to be at a desk
- The same data pipeline that predicts stockouts today can extend to disease-outbreak early warning, staffing optimization, and equipment maintenance scheduling tomorrow — the platform, not just the feature set, is the asset

9. Technology Stack (Reference)

Layer	Technology
Backend API	FastAPI (Python 3.11)
Database	PostgreSQL 16 + PostGIS (geospatial), TimescaleDB in local/dev
Cache / task queue	Redis 7
Background jobs	Celery + Celery Beat (scheduled scoring, predictions, notifications)
ML / forecasting	Prophet, XGBoost, LightGBM, OR-Tools, scikit-learn

Conversational AI	Gemini (Google AI), multilingual
District dashboard	React 18 + TypeScript + Tailwind CSS + i18next (10 languages)
Field worker app	React PWA + Dexie/IndexedDB (offline sync), 7 languages
Maps	Leaflet + OpenStreetMap
Messaging	Meta WhatsApp Cloud API
Cloud infrastructure	Google Cloud Run, Cloud SQL, Compute Engine, Artifact Registry, Secret Manager, Firebase Hosting
Auth	JWT + OTP-based phone login, role-based access control (Field Worker / PHC-CHC Admin / District Officer / State Admin)

10. Brief Description (for the submission form)

PrediCare is an AI-powered district health operating system that predicts operational failures — medicine stockouts, staffing gaps, capacity overruns — at Primary and Community Health Centres 48–72 hours before they occur, and closes the intervention loop in under 5 minutes via one-tap district-officer approval and WhatsApp-based field instructions. It combines time-series forecasting, an optimization-based redistribution engine, and a multilingual conversational AI assistant into a single live dashboard for district officers and an offline-first mobile app for field workers, deployed end-to-end on Google Cloud.

11. Suggested Slide Map (10–12 Slides)

#	Slide	Source section
1	Title + tagline	Header of this doc
2	The problem (2–5 day detection gap)	Section 1
3	The solution, one-sentence pitch	Section 2
4	Live demo walkthrough (screenshots of the 3-min loop)	Section 7
5	How the AI works (module table)	Section 4
6	Who it serves (3 roles)	Section 3
7	Inclusivity — languages, voice, offline	Section 5
8	Why it's deployable (live GCP architecture + links)	Section 6
9	Impact potential / scale story	Section 8
10	Tech stack (compact, logo-style if possible)	Section 9
11	Roadmap / what's next (optional)	—
12	Ask / closing / contact	—

Prepared as source material for the pitch deck — not the deck itself. All figures and architecture details reflect the actual deployed system as of this submission. GitHub repository, deployed links, and live demo credentials available on request from the team.